

The plurivocal university:

Typologizing the diverse voices of a research university on social media

Daniel Chaffey, Ian Wimmer, Nicola Witzig

Motivation and Context

University communication is changing

Challenge of presenting a unified image on social media

The concept of the “plurivocal university”



Motivation and Context

University communication is changing

Challenge of presenting a unified image on social media

The concept of the “plurivocal university”

Motivation and Context

University communication is changing

Challenge of presenting a unified image on social media

The concept of the “plurivocal university”



Research Questions and Objectives

How can the diverse social media accounts of a research university be topologized?

What are the characteristics of these different organizational ‘voices’?

1. Which voices can be **identified** in a plurivocal university, and how can they be **characterized** in terms of reach and activity
2. How do the multiple voices **interact with the official voices on Twitter** through replies, mentions, and retweets?
3. How do the voices of a plurivocal university **interact with each other** through mentions?
4. What **differences in topics and tonality** emerge between voices when interaction via @ mentions with the official voices, and how does this differ from the topics communicated by the official voices?

Research Questions and Objectives

How can the diverse social media accounts of a research university be topologized?

What are the characteristics of these different organizational ‘voices’?

1. Which voices can be **identified** in a plurivocal university, and how can they be **characterized** in terms of reach and activity
2. How do the multiple voices **interact with the official voices on Twitter** through replies, mentions, and retweets?
3. How do the voices of a plurivocal university **interact with each other** through mentions?
4. What **differences in topics and tonality** emerge between voices when interaction via @ mentions with the official voices, and how does this differ from the topics communicated by the official voices?

Methodology

Data and Scope – platforms, time frame, specific university/accounts

Defining ‘voices’ – account type, posting behavior, content

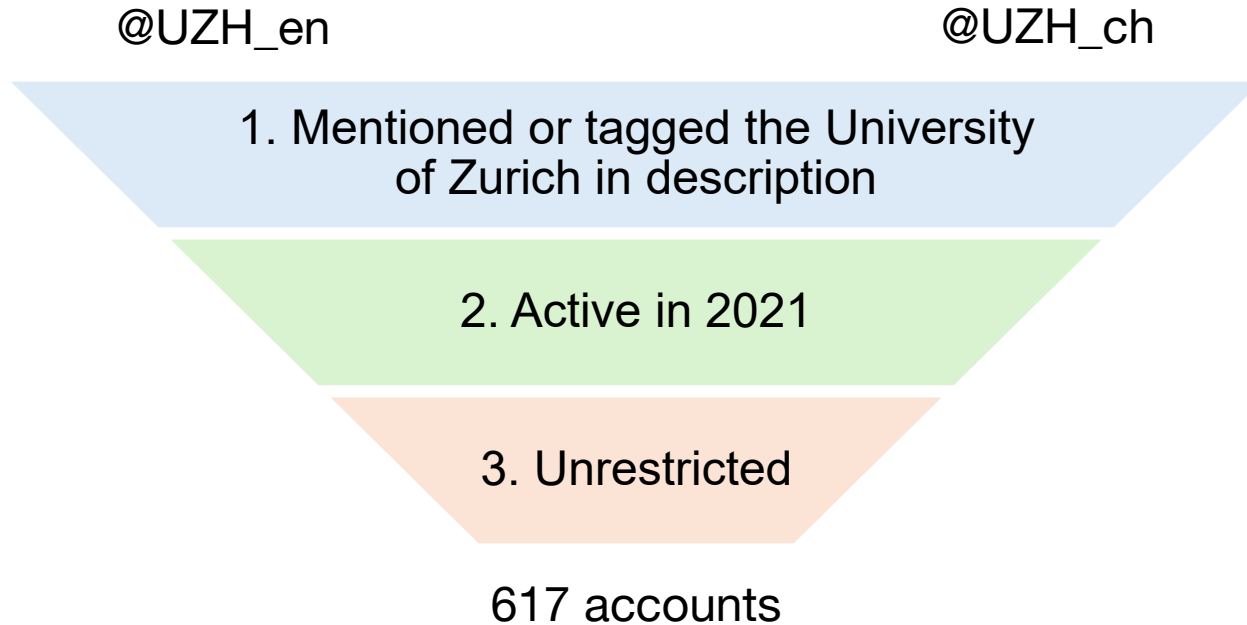
Analysis approach – network analysis, content analysis

Methodology - Data and Scope

X (former Twitter)

Analysis based on the year 2021

Methodology - Data and Scope



Methodology - Defining 'voices'

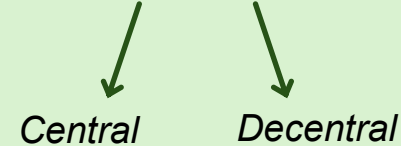
Representation

the permission and legitimacy a voice has
to communicate on behalf of the entire
university



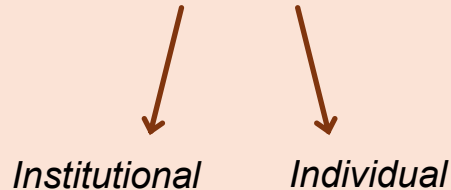
Hierarchy

the position of a voice within the
organizational structure



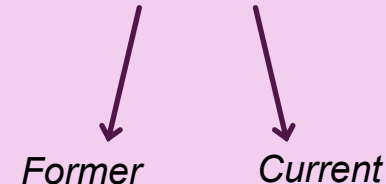
Type

the characteristic of the voice



Affiliation

relationship of the voice with the university



Methodology - Defining 'voices'

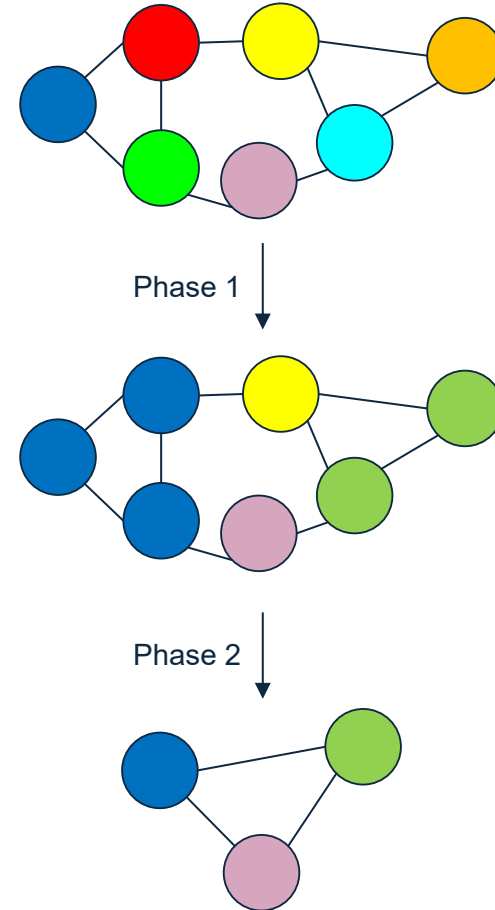
Table 1. Systematization of voices in plurivocal universities.

Voices	Representation	Hierarchy	Type	Affiliation
(1) Official institutional voices (accounts of central communication department)	Yes	Central	Institutional	Current
(2) Official individual voices (official account of president)	Yes	Central	Individual	Current
(3) Central institutional voices (e.g. equal opportunity office, international office)	No	Central	Institutional	Current
(4) Central individual voices (e.g. spokesperson, head of international office)	No	Central	Individual	Current
(5) Decentral institutional voices (e.g. schools, departments, centers)	No	Decentral	Institutional	Current
(6) Decentral individual voices (e.g. professors, students)	No	Decentral	Individual	Current
(7) Former institutional voices (e.g. spinoffs)	No	Decentral	Institutional	Former
(8) Former individual voices (e.g. researchers, alumni)	No	Decentral	Individual	Former

Methodology - Analysis

THE LOUVAIN ALGORITHM

- Highly efficient method for community detection
- Maximise modularity: the density of links inside communities compared to links between communities.



Reproduction Study - End-to-end pipeline

1. Scrape

Follower Harvest

Scraping X
using twscrape
@UZH_en only

2. Filter

Bio Filter

UZH needle
match in name
or bio using
Regex

3. Classify

Code Based

Classify based
on keywords

4. Network

Mention Graph

@-mention
edges Louvain
community

5. Analysis

Content Analysis

Topics – Tonality
Table 2
replication

Methodology - Data and Scope

REPRODUCTION

@UZH_en

1. Mentioned or tagged the University of Zurich in description

2. Active in 2021

3. Unrestricted

859 accounts

ORIGINAL

@UZH_en

@UZH_ch

1. Mentioned or tagged the University of Zurich in description

2. Active in 2021

3. Unrestricted

617 accounts

Results - Voices

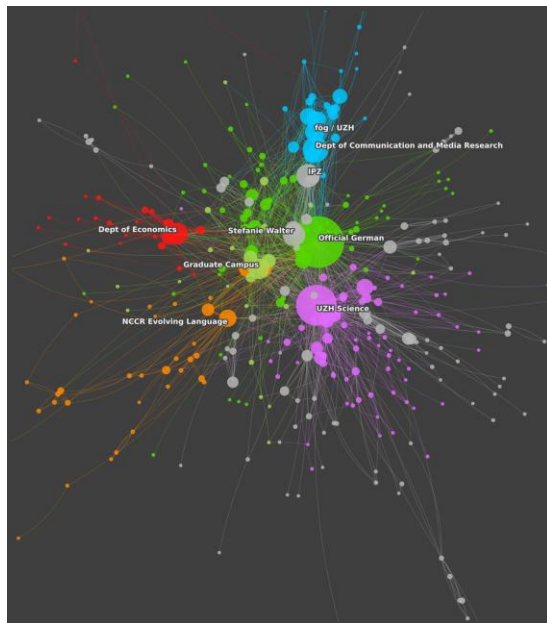
OBSERVATIONS

- Significantly more voices over all
- Strong difference in classification of decentral voices
- Classified some former institutional voices

	REPORDUCTION		ORIGINAL	
	%	n	%	n
Official Institutional	0.0%	1	0.0%	2
Official Individual	0.0%	0	0.0%	0
Central Institutional	0.6%	5	0.6%	4
Central Individual	0.0%	0	1.8%	11
Decentral Institutional	32.4%	278	9.5%	59
Decentral Individual	37.6%	323	71.1%	440
Former Institutional	3.5%	30	0.0%	0
Former Individual	6.4%	55	12.8%	79
Non-Identifiable	19.6%	168	4.2%	26
Total	100.0%	859	100.0%	619

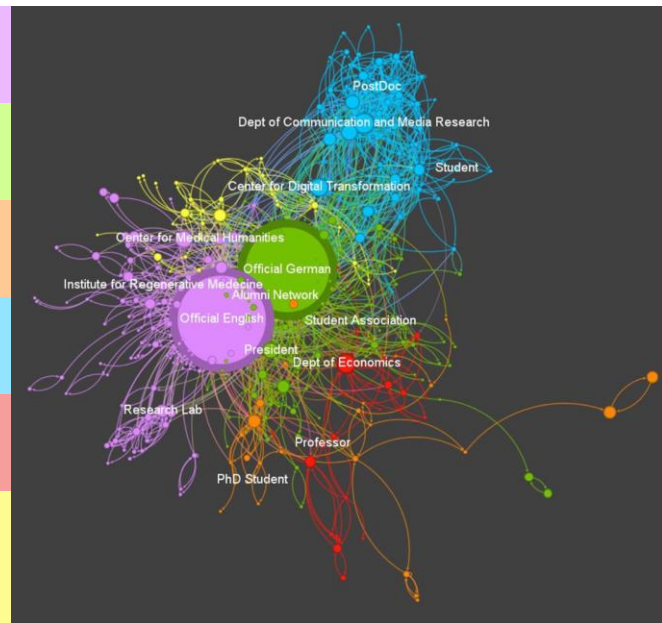
Results - Mention Network

REPRODUCTION



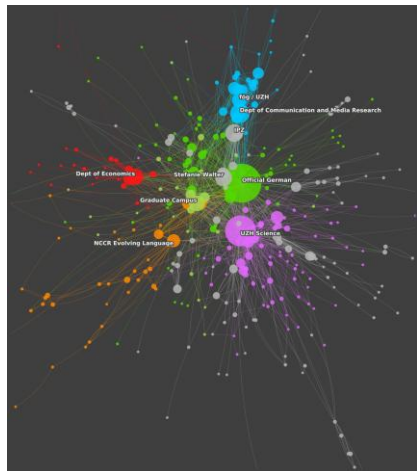
71 (19.0%)	Medicine	106 (28.6%)
66 (17.6%)	Central voices	103 (27.8%)
38 (10.2%)	Linguistics	28 (7.5%)
28 (7.5%)	Social Sciences	56 (15.1%)
27 (7.2%)	Economy	23 (6.5%)
25 (6.7%)	Research Support (←) / Humanities (→)	28 (7.5%)

ORIGINAL

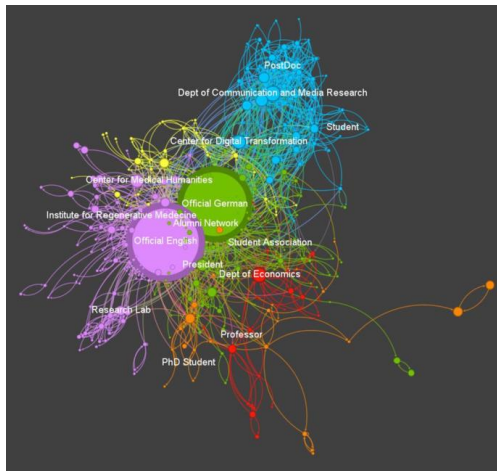


Discussion - Mention Network

REPRODUCTION



ORIGINAL



OBSERVATIONS

Successfully Reproduced:

- Most communities
- Similar structures (e.g. isolation of social science (light blue))

Not Reproduced:

- Order of communities
- Total percentage captured by the largest 6 communities: 68.2% vs. 92.7%

Reproduction Study - Challenges

Twitter/X API changes

- Longer scrape times
- Introduces setup complexity, relying on external scrapers for scheduling
- Reduced dataset, weakens the comparison to the original paper

High manual labor for voice classification

- Relied entirely on keyword coding

Ambiguities in method description

- No mention of Louvain resolution: High resolution => Algorithm splits network into smaller communities
- Mentioned 11 classified communities, later presented only 6 'meaningful' ones

LLM Extension

Research Question

Can an LLM replicate the voice typology automatically at larger scale?

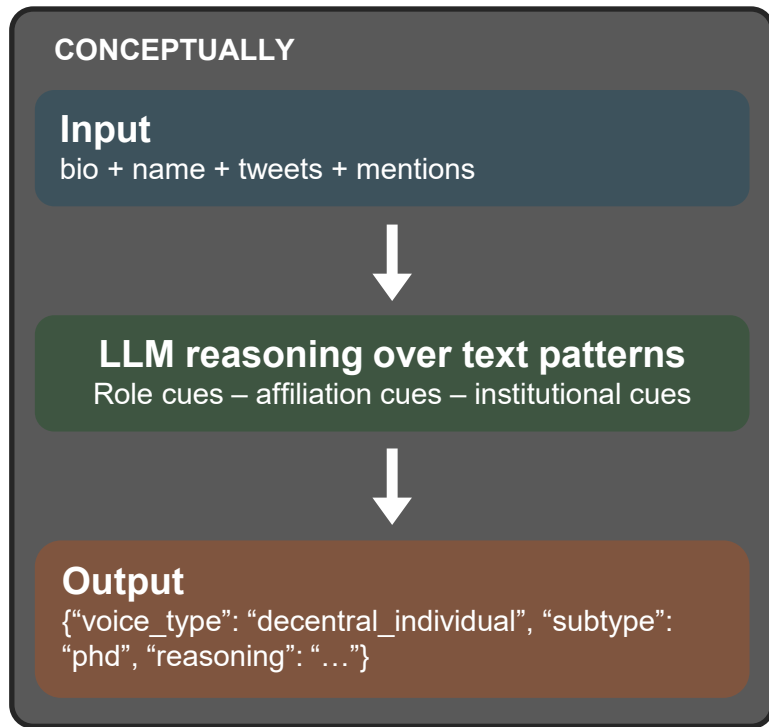
What is a large language model?

Basic idea:

- Neural network trained on massive amounts of text
- Learns language patterns
- It can summarize, extract information, and classify text

For our project

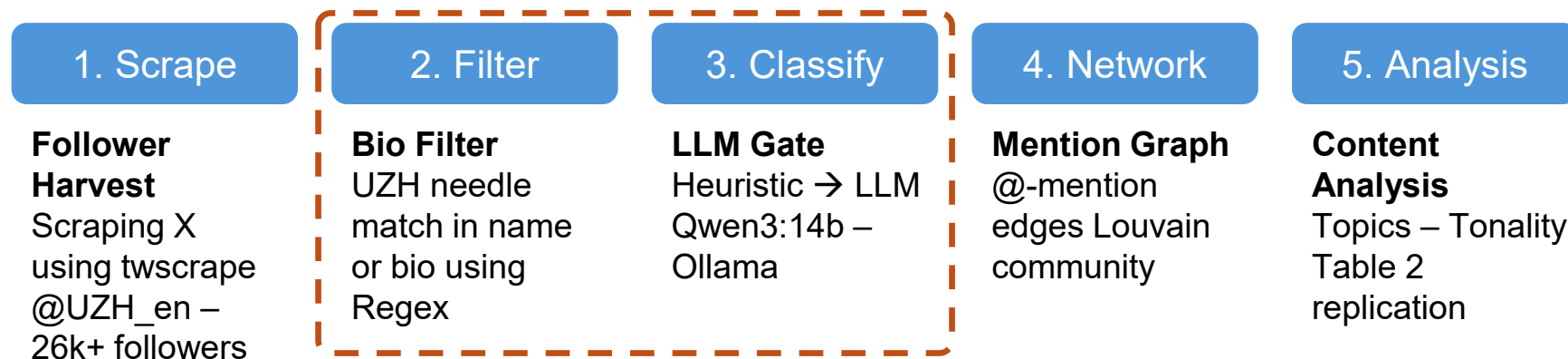
- Used as structured classifier
- Reads bios and tweets, then maps each account to one of the paper's voices.
- Not used for creative text generation in this scenario.



Why do we need an LLM here?



End-to-end pipeline



Two-Tier Pipeline

TIER 1 – HEURISTIC PRE-FILTER

uzh_en / uzh_ch

→ official_institutional

No UZH needle in name / bio

→ non_uzh (*skip LLM*)

UZH bio but role unclear

→ **pass to LLM** (*Tier 2*)

~ 96% of followers are sorted out by heuristic pre-filtering

TIER 2 – LLM CLASSIFICATION

10 voice types – maps exactly to Table 1 of the paper (non_uzh + 8 types + unknown)

Subtypes – professor / postdoc / phd / researcher / student / employee_or_alumni

Checkpoint – progress saved per-account to JSONL so runs can resume after interruption

The Prompt

KEY PROMPT COMPONENTS

Step 1 – UZH affiliation gate

Model must first decide: is this account affiliated with UZH?

If not → **non_uzh** and stop

Otherwise → continue to step 2

Step 2 – Voice type from typology

Full codebook from Table 1 embedded in the system prompt with negative examples (e.g., ETH Zurich $\neq$ UZH, SNSF $\neq$ UZH).

JSON-only output constraint

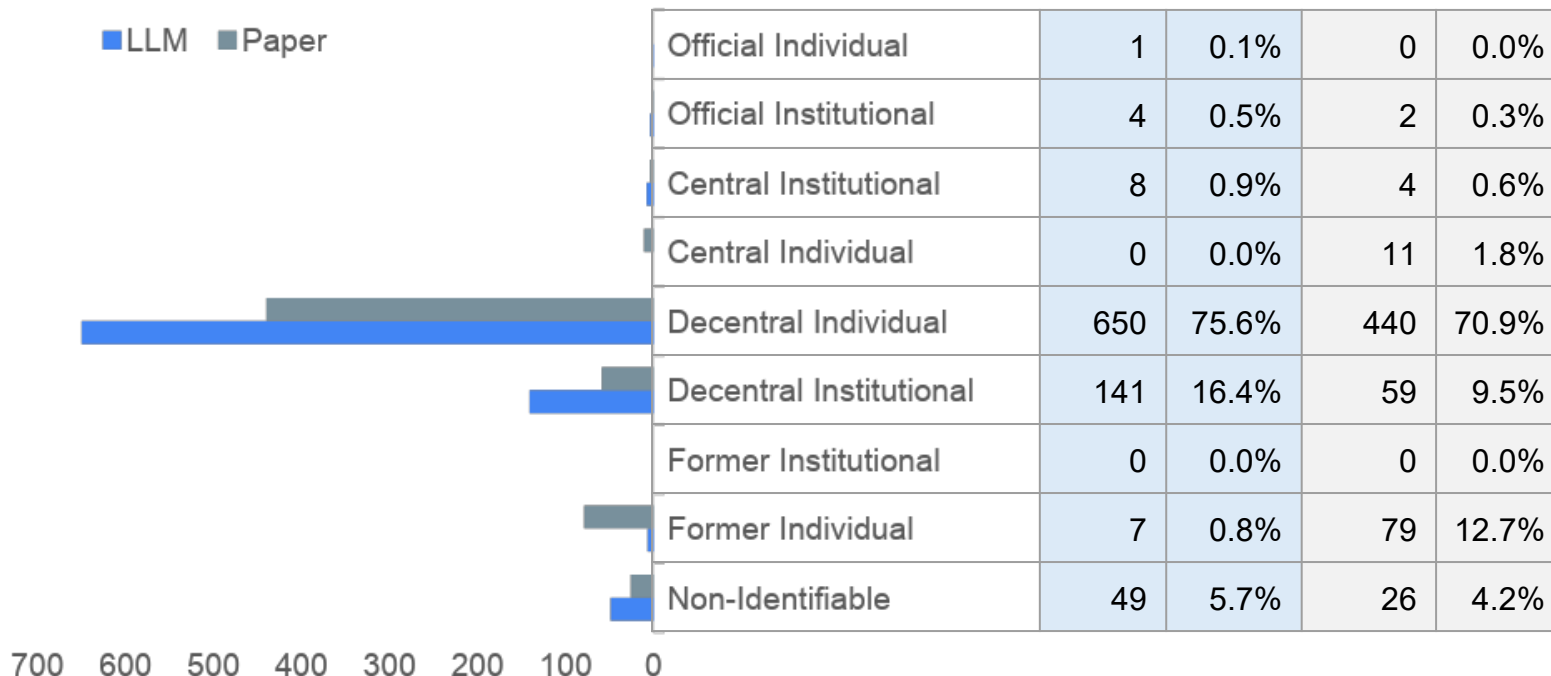
TWEET CONTEXT STRATEGY

8 own tweets – what the account itself posts (discipline, role signals)

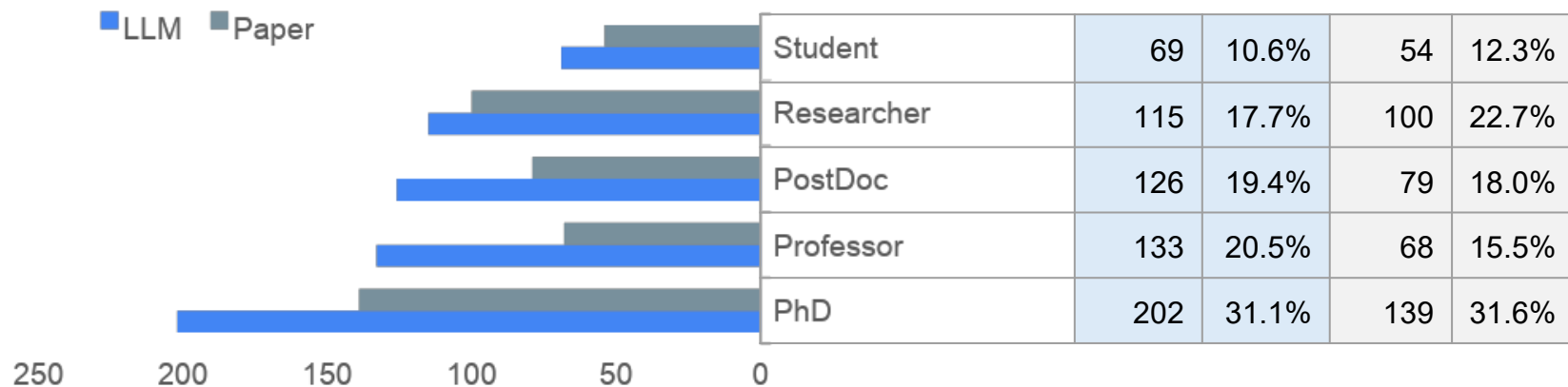
5 mention tweets – how others reference the account (validate role)

Combined with bio, these give the model rich multi-signal context beyond just the profile description.

Voice distribution – Overview

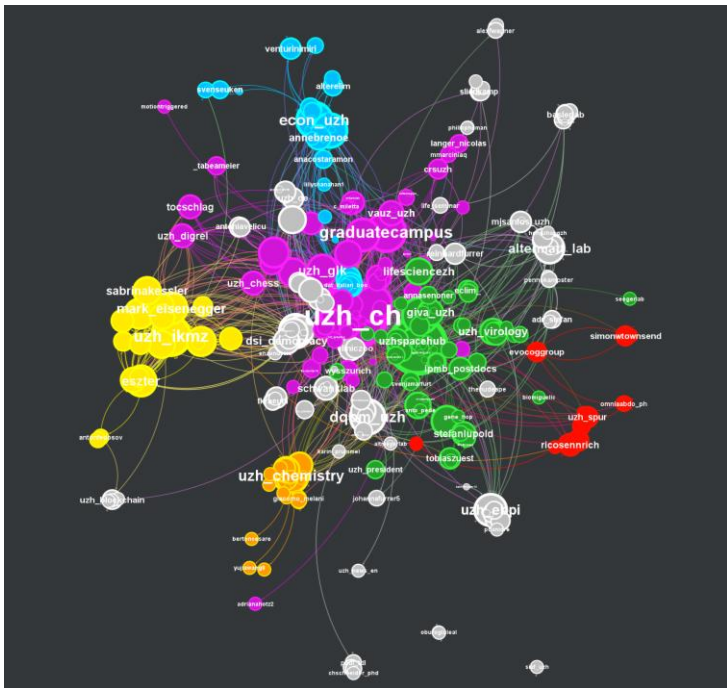


Voice distribution – Decentral Individuals

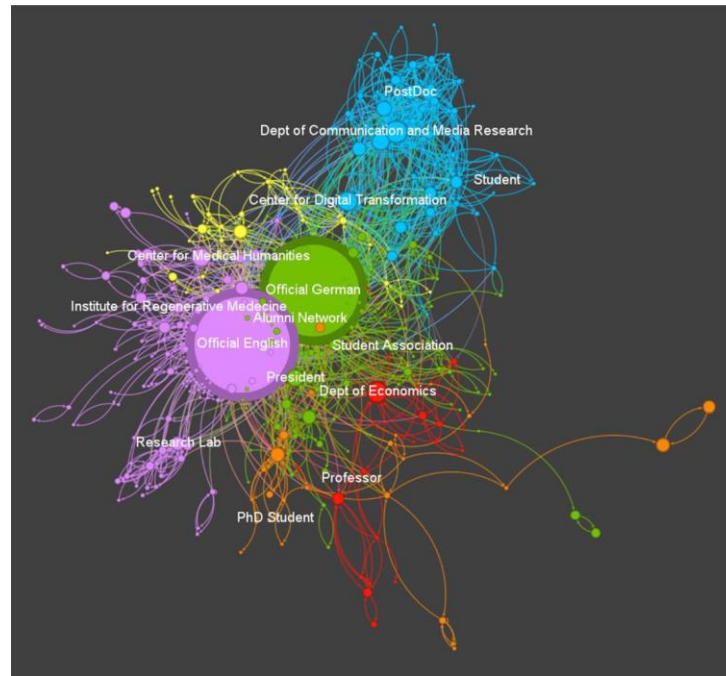


Network Comparison

LLM CLASSIFICATION



ORIGINAL



Limitations and honest caveats

1. No gold-standard validation

No direct comparison possible as original data is not available

Only pattern similarity is showable

2. Different sampling strategy

3. Keyword gate can miss real UZH accounts

4. LLMs can still hallucinate roles

Limitations and honest caveats

1. No gold-standard validation

2. Different sampling strategy

Direct comparison may not be feasible

→ Input data may differ

→ Pre-filtering may differ

3. Keyword gate can miss real UZH accounts

4. LLMs can still hallucinate roles

Limitations and improvement options

1. No gold-standard validation

2. Different sampling strategy

3. Keyword gate can miss real UZH accounts

Heuristic pre-filtering might lead to unintentional flagging UZH-affiliated accounts as non_uzh

4. LLMs can still hallucinate roles

Limitations and honest caveats

1. No gold-standard validation

2. Different sampling strategy

3. Keyword gate can miss real UZH accounts

4. LLMs can still hallucinate roles

Even with strict prompting, the LLM may over-interpret vague bios and infer roles from weak signals in tweets

Final words on LLM extension

The LLM extension is best framed as a plausible, scalable approximation of the original coding – not yet a validated replacement for manual annotation.

Reddit Extension

2024 Reddit r/ethz Analysis Overview

STAGE 1 – DISCOVERY

User Mapping

Enumerates every r/ethz poster/commenter and maps full activity.

Script:

`scrape_user_subreddits_crossref.py`

Output:

CSV with user activity fingerprints.

STAGE 2 – RETRIEVAL

Content Scrape

Fetches full text titles and comments for identified users.

Script:

`scrape_user_posts.py`

Output:

Per-user JSON files (resumable).

STAGE 3 – CLASSIFY

LLM Typology

Two-pass classification into 8 ETH voice categories.

Script:

`classify_content.py`

Output:

CSVs with voice type & evidence.

STAGE 4 – ANALYSIS

Network Analysis

Community detection and bridging analysis (planned).

Script:

`build_interaction_graph.py`
`build_subreddit_graph.py`

Goal:

Identify cross-community bridging agents.

Stage 1 – User Discovery & Mapping

METHOD & PURPOSE

- Exhaustively enumerates users who posted/commented in r/ethz within a date range.
- Maps each user's full Reddit activity across all subreddits via Arctic Shift API.
- Deduplicates usernames and filters out deleted accounts or bots.

EXECUTION SCRIPT

```
scrape_user_subreddits_crossref.py
```

INPUT

Parameters

Target subreddit (ethz) and date range window (after/before YYYY-MM-DD).

OUTPUT

Crossref CSV

User activity fingerprints with packed subreddit:count data for fast loading.

Stage 2 – Per-User Content Retrieval

METHOD & LOGIC

- Retrieves full text (titles, selftext, comments) for users identified in Stage 1.
- Deduplication via seen-ID sets prevents cursor-boundary duplicates during pagination.

EXECUTION SCRIPT

`scrape_user_posts.py`

INPUT

Stage 1 CSV Output

Deduplicated username list with cross-subreddit activity fingerprints.

OUTPUT

Individual JSON Files

Stored in `/user_posts/`. One file per user ensures the pipeline is trivially resumable.

Stage 3 – LLM Voice-Type Classification

TWO-PASS LOGIC (OLLAMA / QWEN3)

- **Pass 1: Extraction** – Isolates verbatim first-person self-id statements using keyword-gated filtering.
- **Pass 2: Classification** – Constrained JSON output maps users to canonical ETH voice typology (Volk et al., 2024).
- **Reduced Hallucination** – Schema-validated grammar ensures only 8 valid categories and subtypes are assigned.
- **Auditability** – Confidence scores (0.0-1.0) and evidence quotes ground every result for manual review.

EXECUTION SCRIPT

```
classify_content.py
```

INPUT

Per-User JSONs

Full post and comment history stored in /user_posts/ from Stage 2.

OUTPUT

Classified CSVs

Detailed results with VOICE_TYPE, evidence quotes, and confidence distributions.

LLM Classification Examples

DECENTRAL_INDIVIDUAL

u/0raid student

"I am currently finishing an MSc in Robotics... At ETH."

u/65289791 phd

"I'm a first year PhD student in chemistry department."

u/Beduines123 researcher

"Currently working at ETH Zurich as a research assistant"

EXTERNAL_INDIVIDUAL

u/Aark95 student

"I am doing the BsC business informatics at WU Vienna"

u/FutureAd6409 phd

"she's a PhD student at Zurich University"

u/0b00000110 other

"ZHAW grad here who has worked in the tech industry..."

FORMER_INDIVIDUAL

u/Ill-Perspective-7190 student

"I've done both my bachelors and masters at ETH"

u/CreativeAd996 phd

"Holding a PhD from ETH and worked in Beijing for some years..."

DECENTRAL_INSTITUTIONAL

u/Fun_Local800 dept/lab

"D-ITET ETH"

u/Smart_Technology_433 dept/lab

"I'm at ITET in my second year"

LLM Classification Statistics

VOICE TYPE	SUBTYPE	N	MEAN POSTS	TOTAL POSTS	CONFIDENCE	SHARE %
decentral_individual	-	1	356	356	0.5	0.06%
decentral_individual	applicant	57	22.04	1'256	0.76	0.22%
decentral_individual	phd	39	230.18	8'977	0.90	1.56%
decentral_individual	student	452	50.13	22'661	0.93	3.94%
external_individual	other	90	205.48	18'493	0.55	3.21%
external_individual	student	327	72.26	23'629	0.78	4.10%
former_individual	-	106	116.80	12'381	0.73	2.15%
unknown	-	3'318	147.04	487'893	0.00	84.75%

Part 1 Methodology – User Interaction Graph Construction

INTERACTION GRAPH LOGIC

- Builds a directed, weighted graph from Reddit reply chains and u/ mentions.
- Identifies communities using Louvain modularity based on interaction density.
- Outputs interactive HTML visualizations and node-level centrality statistics.

EXECUTION SCRIPT

`build_interaction_graph.py`

INPUT

Voice Classifications & Posts

Uses the
`user_voice_classifications.csv`
and the raw JSON posts directory.

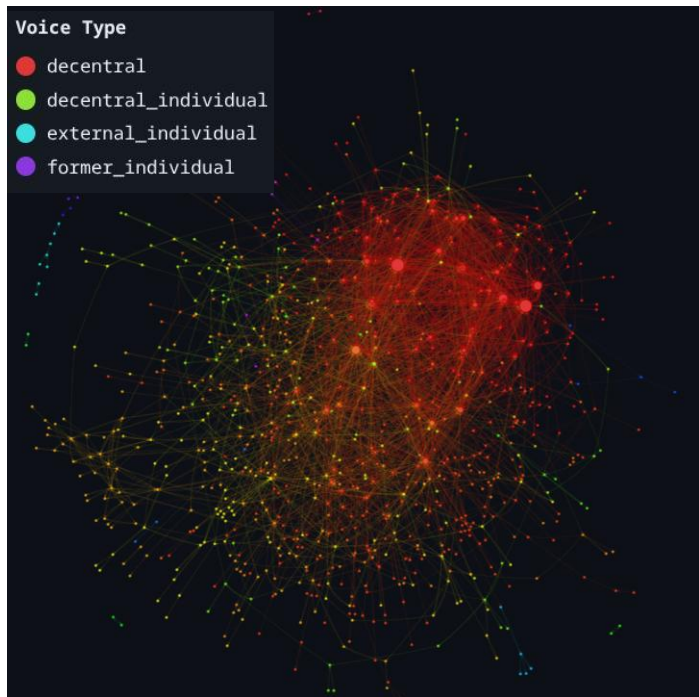
OUTPUT

Interactive Graph (HTML)

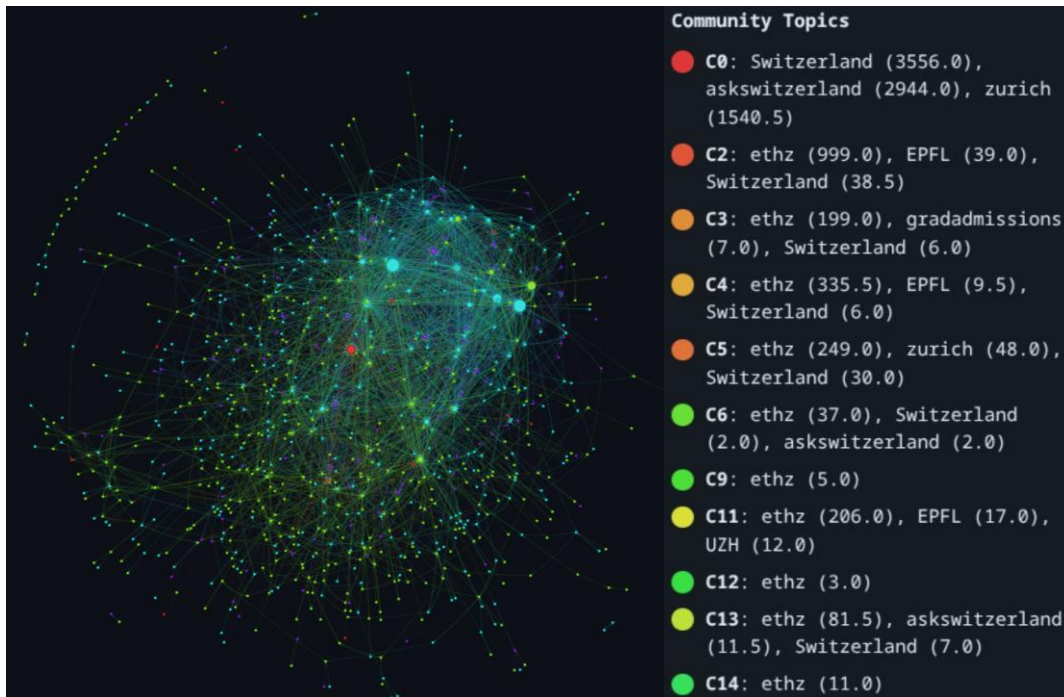
Nodes colored by voice
type/community with Louvain-
partitioned clusters.

Network Visualization Comparisons

COLORED BY COMMUNITY



COLORED BY VOICE TYPE



Part 1 – User Interaction Network Analysis

METHODOLOGY

Nodes: Classified Reddit users.

Edges: Replies, mentions, and co-participation (commenting on the same post)

Algorithms: Louvain (Communities), Assortativity (Homophily)

Graph Size

4'390 Users

575'646 Interactions

1. NETWORK “HAIRBALL”

Modularity (0.2429): Low score indicates a densely connected network. 35.8% of interactions cross community boundaries.

2. LACK OF ECHO CHAMBERS

Assortativity (0.0179): Users have no preference for interacting with their own kind. Students talk to outsiders and applicants.

3. SUPER-SPREADER

Network depends on “power users”. Top user BNI_sp has interaction weight >5'900.

INSIGHTS

Centralization & Bridge Dynamics

The ETH Zurich network is characterized by strong institutional fluidity. The lack of echo chambers suggests a lot of exchange of information between student and external voices.

Stage 4 – Subreddit Co-participation Analysis

CO-PARTICIPATION GRAPH LOGIC

- Builds an undirected, weighted graph where **nodes** represent subreddits with `20 >= --min-users`.
- **Edges** are weighted by the number of classified users active in both subreddits.
- Node attributes include **dominant voice type**, full distribution, and user count.

EXECUTION SCRIPT

```
build_subreddit_graph.py
```

INPUT

User Posts & Classifications

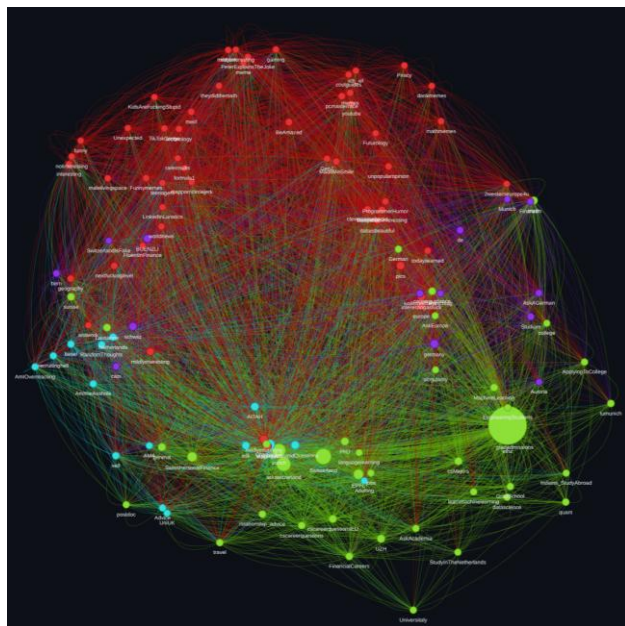
Aggregates .json post history mapped against processed voice classification CSVs.

OUTPUT

Metrics & Visualization

1. **Nodes CSV:** Centrality & Community IDs.
2. **Interactive HTML:** Node-link diagram sized by user density.

Part 2 Methodology – Subreddit Co-participation Graph Construction



1. ACADEMIC & CAREER

Anchored by r/ethz, r/EPFL, and r/PhD. Highest density of current students.

3. VIRAL & ENTERTAINMENT

General Reddit interest like r/mildlyinfuriating and r/meirl.

2. SWISS LIFE & FINANCE

Anchored by r/Switzerland and r/zurich. Focused on local infrastructure.

2. EUROPEAN / DACH

German-speaking world context: r/europe, r/germany, r/de.

FEATURES

Network Statistics & Community Logic

Modularity (0.1870): The subreddit network is even more heavily mixed than the user network. Users from the ETHZ dataset jump across thousands of different subreddits.

Thematic Communities: Despite the low modularity, the Louvain algorithm perfectly clustered the subreddits into 4 distinct thematic categories.

Conclusion – Challenges in Social Network Analysis

KEY LIMITATIONS & FINDINGS

Account Anonymity

Network analysis on Reddit is inherently difficult due to the anonymous nature of every account, complicating user verification and demographic mapping.

Lack of Formal Structure

There is no enforced hierarchy or central authority governing communications, leading to a highly fluid and organic interaction landscape without official university voices.

Institutional Independence

Analysed accounts and subreddits are not backed or managed by the ETH Zurich organization themselves, representing community-driven rather than institutional discourse.

References

[1] Volk, S. C., Vogler, D., Fürst, S., & Schäfer, M. S. (2025). The plurivocal university: Typologizing the diverse voices of a research university on social media. *Public Understanding of Science*, 34(3), 270–290. <https://doi.org/10.1177/09636625241268700>

GitHub page: <https://github.com/nicolawitzig/applied-networks>